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Tutorial: Automate your SP-API Calls Using a Java SDK

Automate your SP-API calls with Login with Amazon (LWA) token exchange and authentication.

This tutorial describes how to generate a Java SDK with Login with Amazon (LWA) token exchange and authentication to build your application seamlessly. You will learn the prerequisites required to build the Java SDK and also view an example using the Selling Partner API (SP-API) for sellers and Swagger Code Generator.

First, take a look at how the authorization process works.

Prerequisites

To complete this tutorial, you need the following prerequisites:

- A hybrid or SP-API app in draft or published state
- Integrated development environment (IDE) software (this walkthrough uses Eclipse IDE on Windows OS)

You must register your application before it can connect to the Selling Partner API, and it must be authorized by a selling partner. If you do not have a hybrid or SP-API app, follow the steps to [register as a developer](#), [register your application](#), and [Authorizing Selling Partner API applications](#). Then, return to this tutorial.

Next, set up your workspace for the tutorial.

Step 1. Set up your workspace

You must download the required software packages and the [Swagger Code Generator](#). Then, clone the [SP-API models repository](#).

1. On your local drive, create a directory for this project and name it **SwaggerToCL**.
2. Download the following tools and make them available in your `$PATH`.
 1. [Java 17 or newer](#) or [OpenJDK 17 or newer](#)
 2. [Apache Maven 3.9 or greater](#)
 3. [GNU Wget 1.20 or greater](#)

3. Run the following command to download the Swagger Code Generator:

Bash

```
wget
https://repo1.maven.org/maven2/io/swagger/codegen/v3/swagger-
```

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codegen-cli/3.0.61/swagger-codegen-cli-3.0.61.jar

- Copy `swagger-codegen-cli.jar` into your local directory `C:\SwaggerToCL`.
- In GitHub, go to <https://github.com/amzn/selling-partner-api-models/tree/main/models> and use the following command to clone the `selling-partner-api-models` repository to your local directory `C:\SwaggerToCL`.

Bash

```
git clone https://github.com/amzn/selling-partner-api-models
```

- Navigate to the `selling-partner-api-models\models` folder in your local copy of the repository and copy all JSON files from the models subfolders into `C:\SwaggerToCL`.

The `selling-partner-api-models` repository has two directories:

- [models directory](#) - contains all the currently available Selling Partner API models.

**Note**

Developers use [Swagger Codegen](#) to generate client libraries from these models. Swagger CodeGen is an open source project which enables automatic generation of API client libraries (SDK generation), server stubs, and documentation from an OpenAPI specification. For more information, refer to [Swagger Codegen](#).

- [clients directory](#) - contains a Java library named `sellingpartner-api-aa-java` and a C# library named `sellingpartner-api-aa-csharp` with Mustache templates for use with `swagger-codegen` to generate client libraries with authentication and authorization functionality included. The `sellingpartner-api-documents-helper-java` folder contains helper classes for encrypted documents. The `sample-code` folder contains java code examples for the [Restricted Data Token \(RDT\)](#).

Now that you have completed the required setup, the next step is to generate the Java SDK with the authentication and authorization classes provided in the `sellingpartner-api-aa-java` folder.

Step 2. Generate Java SDK for SP-APIs

In this step, you generate the SDK against the templates in the folder `selling-partner-api-models\clients\sellingpartner-api-aa-java` in your local copy of the repository. This folder contains an authorization and authentication library, along with customized templates for the Swagger Code Generator, and a config file to generate the correct version of the SDK.

To generate the Java SDK, you need to run the following command for each JSON file in your local directory:

Bash

```
java -jar C:\SwaggerToCL\swagger-codegen-cli.jar generate \
-i C:\SwaggerToCL\[name of model].json \
-l java -t [path to selling-partner-api-models\clients\sellingpartner-api-aa-java folder]\resources\swagger-codegen\templates\ \
-o C:\SwaggerToCL\[name of client library] \
```

```
-c [path to selling-partner-api-models\clients\sellingpartner-api-aa-
java folder]\resources\java\config.json
```

For example, run the following command to generate Java code for **sellers.json**.

Bash

```
java -jar C:\SwaggerToCL\swagger-codegen-cli.jar generate \
-i C:\SwaggerToCL\Sellers.json \
-l java -t [path to selling-partner-api-
models\clients\sellingpartner-api-aa-java folder]\resources\swagger-
codegen\templates\ \
-o C:\SwaggerToCL\Sellers_JavaCL \
-c [path to selling-partner-api-models\clients\sellingpartner-api-aa-
java folder]\resources\java\config.json
```

Repeat this step to generate SDKs for each of the SP-APIs available in the [models directory](#).

The generated SDK includes the classes for configuring your LWA credentials and uses them to exchange LWA tokens.

In the following steps, **x.x** refers to the latest version of the AA library. Refer to the [readme](#) to get the latest version.

Step 3. Build the Authentication/Authorization Library (AA)

After you generate the SDK, build the AA library.

1. Navigate to the **selling-partner-api-models\clients\sellingpartner-api-aa-java** folder of your local copy of the repository and run `mvn package`. This generates a folder named `target`. In this folder is a JAR file named **sellingpartnerapi-aa-java-x.x-jar-with-dependencies.jar** (or similarly named) and all of the required dependencies.

2. Run the following command to install the JAR file in your local Maven repository.

mvn

```
mvn install:install-file -Dfile=[path to JAR file in "target"
folder] -DgroupId=com.amazon.sellingpartnerapi -
DartifactId=sellingpartnerapi-aa-java -Dversion=x.x -
Dpackaging=jar
```

You can find the actual `groupId`, `artifactId`, and `version` values near the top of the `pom.xml` file in the **selling-partner-api-models\clients\sellingpartner-api-aa-java** folder.

Example:

mvn

```
mvn install:install-file -Dfile=C:\SwaggerToCL\sellingpartner-
api-aa-java\target\sellingpartnerapi-aa-java-x.x.jar -
DgroupId=com.amazon.sellingpartnerapi -
DartifactId=sellingpartnerapi-aa-java -Dversion=x.x -
Dpackaging=jar
```

mvn

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```
mvn install:install-file -Dfile=C:\SwaggerToCL\sellingpartner-api-aa-java\target\sellingpartnerapi-aa-java-x.x-jar-with-dependencies.jar -DgroupId=com.amazon.sellingpartnerapi -DartifactId=sellingpartnerapi-aa-java -Dversion=x.x -Dpackaging=jar
```

Step 4. Add the library as a dependency of the SDK

To add the library as a dependency, you need to edit the `pom.xml` file.

1. In your local directory, navigate to the generated Seller API SDK and open the `pom.xml` file.
2. Add the following code as a dependency in the file.

XML

```
<dependency>
<groupId>com.amazon.sellingpartnerapi</groupId>
<artifactId>sellingpartnerapi-aa-java</artifactId>
<version>x.x</version>
</dependency>
```

 Note

`x.x` refers to the latest version of the AA library. Refer to the [readme](#) to get the latest version.

3. Save and close the file.

The next step is to write the actual application and connect to the SP-API using the generated Java SDK.

Step 5. Connect to the Selling Partner API using the generated Java SDK

In this step, you connect to the Selling Partner API using the Java SDK generated in the preceding steps. These steps use [Eclipse IDE](#) to import the generated SDK and connect to the Selling Partner API, but you can use your preferred IDE.

1. In Eclipse IDE, on the **File** menu, choose **Open Projects from File System** and then import the generated Java SDK.
2. In `src/main/java` add a `class` to write your code.
3. Enter code in the added `class` to configure the following instance:
 1. LWA credentials (refer to [Configure your LWA credentials](#))

 Note

If the class path doesn't recognize the classes in Eclipse, you'll need to add the `JAR` files that were created in the `target` folder, which is located in the generated SDK folder.

4. Create an instance of the generated SDK API and call the operations. For an example, refer to the following sample code for connecting to the Selling Partner API for Sellers (provided that you generated the SDK for this API).

Java

Amazon Selling Partner API Guard Implementation Guide

VAT Calculation Service

Amazon Seller Data Access

Best Practices

A+ CONTENT API

A+ Content API

A+ Content Examples

AMAZON WAREHOUSING AND DISTRIBUTION API

Amazon Warehousing and Distribution API

```
import io.swagger.client.*;
import io.swagger.client.auth.*;
import io.swagger.client.model.*;
import io.swagger.client.api.SellersApi;
import java.io.File;
import java.util.*;
import
com.amazon.SellingPartnerAPIAA.LWAAuthorizationCredentials;
import com.amazon.SellingPartnerAPIAA.LWAException;
import static
com.amazon.SellingPartnerAPIAA.ScopeConstants.SCOPE_NOTIFICATIONS_API;
// for grantless operations (Notifications API)
import static
com.amazon.SellingPartnerAPIAA.ScopeConstants.SCOPE_MIGRATION_API;
// for grantless operations (Authorization API)

public class SellersApiDemo {

    public static void main(String[] args) {

        //Configure your LWA credentials :
        LWAAuthorizationCredentials lwaAuthorizationCredentials =
        LWAAuthorizationCredentials.builder()
            .clientId("amzn1.application-
            *****")

        .clientSecret("*****")

        .refreshToken("Atzr|*****")
            .endpoint("https://api.amazon.com/auth/o2/token")
            .build();

        //For Grantless operations (Authorization API and
        Notifications API), use following code :
        /* LWAAuthorizationCredentials
        lwaAuthorizationCredentials =
        LWAAuthorizationCredentials.builder()
            .clientId("amzn1.application-
            *****")

        .clientSecret("*****")
            .withScopes(SCOPE_NOTIFICATIONS_API,
        SCOPE_MIGRATION_API)
            .endpoint("https://api.amazon.com/auth/o2/token")
            .build(); */

        //Create an instance of the Sellers API and call an
        operation :
        SellersApi sellersApi = new SellersApi.Builder()

        .lwaAuthorizationCredentials(lwaAuthorizationCredentials)
            .endpoint("https://sellingpartnerapi-
            na.amazon.com") // use Sandbox URL here if you would like to test
            your applications without affecting production data.
            .build();

        try {
            GetMarketplaceParticipationsResponse result =
            sellersApi.getMarketplaceParticipations();
            System.out.println(result);
        } catch (ApiException e) {
            System.out.println("Exception when calling
            SellersApi#getMarketplaceParticipations");
            e.printStackTrace();
        }

        You can follow similar steps for other SP-APIs.
        catch (LWAException e) {
```

Now that you have run the code, on the main class, you now have a list of marketplaces that your authorized seller can sell in.

Amazon Warehousing and Distribution API v2024-05-09 Reference

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App Integrations API >

Updated 18 days ago

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